

We removed Pedestrian Accidents and Pedestrian Injuries as they only had 7 years of data.

We accidentally used the target variable as a predictor.

Mecklenburg County Traffic Fatality Risk Classification

Overview

A binary classification model was built to predict whether a given year in Mecklenburg County would be a "high fatality" or "low fatality" year for traffic accidents. Years were labeled using the median of Traffic Accident Fatalities across all years in the dataset (1981–2024). Years above the median were labeled 1 (high risk) and years at or below were labeled 0 (low risk), producing a near-perfectly balanced dataset of 22 high-risk and 21 low-risk years.

Data & Features

Data was sourced from two datasets: the NC Transportation LINC dataset and Mecklenburg County population estimates. All accident-based features were normalized per 100,000 residents to account for population growth over the 43-year span. Two features — Pedestrian Accidents and Pedestrians Injured — were excluded due to only being available for 7 years (2018–2024), which would have reduced the modeling dataset to an unusable size. Traffic Accident Fatalities per capita was also excluded as a feature despite being available, because it was used to construct the target label and its inclusion constituted target leakage — the model would predict fatality risk almost entirely based on the death count itself rather than the behavioral factors driving those deaths.

The final five features used were:

- Alcohol Related Accidents per capita
 - Accidents Caused by Exceeding Legal Speed per capita
 - Motorcyclists Injured per capita
 - Bicyclists Injured per capita
 - Population Growth Percentage
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Model Building Process

Three classification methods were evaluated. All models were assessed using 5-fold Stratified Cross-Validation with shuffling to ensure balanced class representation in each fold. Performance was measured across three metrics — Accuracy, F1 Score, and ROC-AUC — to capture both overall correctness and the balance between precision and recall.

Logistic Regression was used as the interpretable baseline. A regularization strength of $C=0.1$ was chosen to prevent overfitting given the small sample size, and features were standardized so coefficients could be compared directly. However, Logistic Regression performed the weakest of the three models and produced counterintuitive negative coefficients for speeding and alcohol-related accidents. This is attributed to a strong historical trend in the data — speeding and alcohol enforcement increased significantly from the 1980s onward while fatality rates simultaneously declined due to improved vehicle safety and trauma care. Logistic Regression could not separate this historical trend from the underlying behavioral signal, making its coefficients unreliable for interpretation in this context.

Random Forest was included to capture potential non-linear relationships between features. Trees were kept shallow (max depth of 3) and required a minimum of 3 samples per leaf to avoid overfitting on 43 rows. 300 trees were used for stable importance estimates. Random Forest outperformed Logistic Regression on all metrics and produced feature importances that align well with domain knowledge, making it the preferred model for this analysis.

K-Nearest Neighbors (KNN) was included as a non-parametric contrast to the other two methods. $K=7$ was chosen because it is odd (avoiding tie votes) and approximates the square root of the sample size, a standard heuristic for small datasets. Features were standardized since KNN is distance-based and sensitive to scale. KNN achieved the highest accuracy and ROC-AUC of the three models, confirming that the relationship between these behavioral features and fatality risk is non-linear in nature.

Results

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.608	0.604	0.615
Random Forest	0.678	0.672	0.675
KNN (k=7)	0.697	0.617	0.700

All three models outperformed the 50% random baseline expected for a balanced binary classification problem. KNN achieved the highest accuracy (69.7%) and ROC-AUC (0.700), while Random Forest produced the strongest F1 score (0.672) and the most interpretable feature importances. Logistic Regression was the weakest performer across all metrics, attributed to its assumption of linear relationships between predictors and the outcome — an assumption that does not hold well given the historical trends embedded in this dataset.

Random Forest is selected as the primary model for interpretation due to its balance of performance and interpretability.

Important Predictors

Accidents Caused by Exceeding Legal Speed per capita was the most important feature in the Random Forest model with a Gini importance of 0.28. Years with higher per-capita speeding accidents are the strongest behavioral predictor of a high fatality year, which aligns with the expectation that speed is a major contributor to crash severity.

Alcohol Related Accidents per capita was the second most important feature with a Gini importance of 0.26. This is nearly as influential as speeding, supporting the well-established relationship between impaired driving and fatal crash risk.

Motorcyclists Injured per capita ranked third with a Gini importance of 0.20. Motorcyclists are among the most vulnerable road users, and years with elevated motorcyclist injury rates are meaningfully associated with higher overall fatality risk.

Bicyclists Injured per capita contributed moderately with a Gini importance of 0.16. While less dominant than the top predictors, bicyclist injury rates add additional signal about vulnerable road user exposure in a given year.

Population Growth Percentage was the weakest predictor with a Gini importance of 0.10. This suggests that how fast Mecklenburg County was growing in a given year had limited bearing on fatality risk once per-capita normalization was applied to the other features.

Logistic Regression Coefficients

While Logistic Regression is not the recommended model for this analysis, its coefficients are reported for completeness. Assuming all other predictors remain constant:

- A one unit increase in Accidents Caused by Exceeding Legal Speed per capita decreases the log-odds of a high risk year by 0.17
- A one unit increase in Alcohol Related Accidents per capita decreases the log-odds of a high risk year by 0.10
- A one unit increase in Motorcyclists Injured per capita decreases the log-odds of a high risk year by 0.05
- A one unit increase in Population Growth Percentage increases the log-odds of a high risk year by 0.03
- A one unit increase in Bicyclists Injured per capita increases the log-odds of a high risk year by 0.02

The negative direction of the top coefficients is counterintuitive and is attributed to the historical trend described above — increased enforcement and reporting of speeding and alcohol incidents coincided with decades of declining fatality rates, causing the model to associate high counts of these incidents with lower fatality risk. This limitation reinforces the decision to rely on Random Forest as the primary model.

Conclusion

The Random Forest classifier is the recommended model for predicting high fatality years in Mecklenburg County, achieving 67.8% accuracy and a ROC-AUC of 0.675 using only behavioral and demographic indicators. Speeding and alcohol-related accidents per capita are the strongest predictors of a dangerous year, followed by motorcyclist and bicyclist injury rates. Future work could explore adding pedestrian data once longer historical records become available, incorporating weather or infrastructure variables, or applying a time-aware labeling approach to better account for the historical trends present in this dataset.

Our goal when we started making the classification model was to develop a binary classification model to predict high versus low traffic fatality risk years in Mecklenburg County, North Carolina. We used two datasets to create our model. The first was transportation data spanning from 1970-2024, containing features like alcohol-related accidents, speed violations, motorcyclist accidents, and pedestrian accidents. The second was population data in Mecklenburg from the same time period. This lets us factor the importance of different types of accidents and the effect of population counts on raw accident counts.